

A COUNTEREXAMPLE TO THE AGGREGATE-CONDITION FORM OF KARP–SITNIK CONJECTURE 1

ABSTRACT. We consider the nonredundant aggregate-condition form of Conjecture 1 of Karp and Sitnik, in which the componentwise conditions $b_i > a_i > 0$ are replaced by $\sum_i (b_i - a_i) > 0$ while positivity is retained. We give the exact counterexample

$${}_3F_2\left(\begin{matrix} 2, 2, 2 \\ 1, 5 \end{matrix}; -\frac{1}{2}\right) = 1032 - 2544 \log \frac{3}{2} < \frac{25}{49}.$$

Thus the aggregate condition is insufficient already for $q = 2$, $\sigma = 2$, positive integer parameters, and $\prod_i a_i/b_i < 1$. A quadratic expansion also gives an infinite family of counterexamples for every $\sigma > 0$.

Let $q \geq 1$. For $A = (a_1, \dots, a_q)$ and $B = (b_1, \dots, b_q)$ in $(0, \infty)^q$, set

$$H_{\sigma, A, B}(x) := {}_{q+1}F_q\left(\begin{matrix} \sigma, a_1, \dots, a_q \\ b_1, \dots, b_q \end{matrix}; -x\right), \quad \mu(A, B) := \prod_{i=1}^q \frac{a_i}{b_i}.$$

Karp and Sitnik proved that for $b_i > a_i > 0$, $\sigma \geq 1$ and $x > -1$,

$$(1) \quad (1 + \mu(A, B)x)^{-\sigma} < H_{\sigma, A, B}(x),$$

[1, Theorem 3]. Their Conjecture 1 reads, in full, “Theorem 3 is true for all $\sigma > 0$ and $\sum_{i=1}^q (b_i - a_i) > 0$ ” [1, Conjecture 1]. We read the condition

$$(2) \quad \sum_{i=1}^q (b_i - a_i) > 0$$

as replacing the componentwise inequalities $b_i > a_i$ of Theorem 3, all parameters remaining positive. Three features of [1] support this reading. First, (2) is automatic under the strict inequalities $b_i > a_i$ of Theorem 3, so on a reading that retains them it would be an idle clause. Second, Conjecture 2 of [1] lists “ $0 < \sigma \leq \min(a_1, a_2, \dots, a_q)$, $\sum_{i=1}^q (b_i - a_i) > 0$ and $x > 0$ ” as its complete parameter hypothesis, with no componentwise clause; in [1] the aggregate condition is thus used as a substitute for $b_i > a_i$ and not as an addition to it. Third, in the discussion following Theorem 3 the authors record that (1) for $b_i \geq a_i > 0$, $\sigma > 0$ and $x > 0$ is already due to Luke, and that under the componentwise hypotheses the only case left open is $0 < \sigma < 1$ with $-1 < x < 0$, which they conjecture there in prose; a componentwise Conjecture 1 would therefore restate a theorem they credit to Luke on $x > 0$ and that prose conjecture on $-1 < x < 0$.

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These three features do not exclude one intermediate reading, and we do not claim that they do: one may take Conjecture 1 to assert (1) under $b_i \geq a_i > 0$ together with (2), a hypothesis under which the aggregate clause is not idle, since it then excludes only $A = B$, and which the non-strict form of Luke's result recorded in [1] makes salient. Our example does not bear on that intermediate reading: it has $b_1 = 1$ and $a_1 = 2$, so even $b_1 \geq a_1$ fails, and indeed no violation of (1) for $x > 0$ can occur when $b_i \geq a_i > 0$, precisely because of the bound credited to Luke. What Theorem 1 refutes is the reading stated above, on which (2) replaces the componentwise inequalities altogether; on the intermediate reading the question remains open.

The two halves of Conjecture 1 have fared differently. The relaxation of $\sigma \geq 1$ to $\sigma > 0$ is established on ranges controlled by majorization, in [3, Theorem 5] and, on a wider Meijer- G range, in [2], where the resulting lower bound is described as settling "Conjecture 1 of [1] on that range", with the remark that neither bound obtained there "addresses the broader conjectural condition $\sum_i (b_i - a_i) > 0$ " [2, discussion following Proposition 3.3]. It is that broader condition, the replacement of $b_i > a_i > 0$ by (2), which fails. The counterexample below is taken at $x = 1/2$, which lies in the range $x > -1$ of Theorem 3 as well as in the range $x > 0$ of the bound credited to Luke in [1], so the refutation does not depend on which of the two x -ranges is attached to Conjecture 1.

Theorem 1. *The aggregate-condition form of Conjecture 1 of Karp and Sitnik is false. In fact,*

$$H_{2,(2,2),(1,5)}\left(\frac{1}{2}\right) < \left(1 + \frac{4}{5} \cdot \frac{1}{2}\right)^{-2} = \frac{25}{49}.$$

Proof. The parameters satisfy

$$(1 - 2) + (5 - 2) = 2 > 0, \quad \mu((2, 2), (1, 5)) = \frac{4}{5}.$$

The coefficient of z^n in ${}_3F_2(2, 2, 2; 1, 5; z)$ is

$$\frac{(2)_n^3}{(1)_n(5)_n n!} = \frac{24(n+1)^2}{(n+2)(n+3)(n+4)} = \frac{12}{n+2} - \frac{96}{n+3} + \frac{108}{n+4}.$$

For $0 < |z| < 1$ and an integer $r \geq 1$,

$$\sum_{n=0}^{\infty} \frac{z^n}{n+r} = z^{-r} \left(-\log(1-z) - \sum_{j=1}^{r-1} \frac{z^j}{j} \right).$$

Substituting $z = -1/2$ gives

$$(3) \quad {}_3F_2\left(\begin{matrix} 2, 2, 2 \\ 1, 5 \end{matrix}; -\frac{1}{2}\right) = 1032 - 2544 \log \frac{3}{2}.$$

Moreover,

$$\log \frac{3}{2} = 2 \operatorname{artanh} \frac{1}{5} = 2 \sum_{k=0}^{\infty} \frac{1}{(2k+1)5^{2k+1}} > 2 \left(\frac{1}{5} + \frac{1}{375} + \frac{1}{15625} \right) = \frac{19006}{46875}.$$

Therefore

$$1032 - 2544 \log \frac{3}{2} < \frac{7912}{15625} < \frac{25}{49},$$

where the last inequality follows from

$$25 \cdot 15625 - 49 \cdot 7912 = 2937 > 0.$$

□

The value $q = 2$ is minimal for this phenomenon, since for $q = 1$ condition (2) is exactly $b_1 > a_1$.

Proposition 2. *Let $A, B \in (0, \infty)^q$ and $\sigma > 0$, and define*

$$P_A := \prod_{i=1}^q \left(1 + \frac{1}{a_i}\right), \quad P_B := \prod_{i=1}^q \left(1 + \frac{1}{b_i}\right).$$

If $P_B > P_A$, then there exists $\varepsilon > 0$ such that

$$H_{\sigma,A,B}(x) < (1 + \mu(A, B)x)^{-\sigma} \quad (0 < x < \varepsilon).$$

Proof. As $x \rightarrow 0$,

$$\begin{aligned} H_{\sigma,A,B}(x) &= 1 - \sigma\mu x + \frac{\sigma(\sigma+1)}{2}\mu^2 \frac{P_A}{P_B} x^2 + O(x^3), \\ (1 + \mu x)^{-\sigma} &= 1 - \sigma\mu x + \frac{\sigma(\sigma+1)}{2}\mu^2 x^2 + O(x^3), \end{aligned}$$

where $\mu = \mu(A, B)$. Hence

$$H_{\sigma,A,B}(x) - (1 + \mu x)^{-\sigma} = \frac{\sigma(\sigma+1)}{2}\mu^2 \left(\frac{P_A}{P_B} - 1\right) x^2 + O(x^3),$$

whose leading coefficient is negative when $P_B > P_A$. The difference is therefore negative for all sufficiently small $x > 0$. □

Corollary 3. *For every integer $m \geq 2$ and every $\sigma > 0$, let*

$$A = (m, m), \quad B = (1, m^2 + 1).$$

Then (2) holds and $\mu(A, B) < 1$, but (1) fails for all sufficiently small $x > 0$.

Proof. Here

$$\sum_{i=1}^2 (b_i - a_i) = m^2 - 2m + 2 > 0, \quad \mu(A, B) = \frac{m^2}{m^2 + 1} < 1.$$

Also

$$P_A = \left(1 + \frac{1}{m}\right)^2, \quad P_B = 2 \left(1 + \frac{1}{m^2 + 1}\right).$$

For $m = 2$,

$$P_A = \frac{9}{4} < \frac{12}{5} = P_B;$$

for $m \geq 3$,

$$P_A \leq \frac{16}{9} < 2 < P_B.$$

Proposition 2 applies. □

Karp and Prilepkina later proved (1) for every $\sigma > 0$ under weak supermajorization [3, Theorem 5]; there $B \prec^W A$ means that, both vectors being arranged increasingly, $\sum_{i=1}^k a_i \leq \sum_{i=1}^k b_i$ for $k = 1, \dots, q$. The present example is not covered: after increasing rearrangement the case $k = 1$ of this condition would require $a_1 \leq b_1$, that is, $2 \leq 1$. Its case $k = q$, on the other hand, is (2) with equality allowed, so Theorem 1 shows that the hypothesis $B \prec^W A$ in [3, Theorem 5] cannot be weakened to that final partial sum alone.

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